Large-Scale High-Resolution Essential Urban Land Cover Category Mapping Using a Semantic-Augmented and Noise-Tolerant Approach

Ziming Li, Han Wang, Yadian Wang, Gang Chen, Jing M. Chen, and Bin Chen

***Abstract*—High-resolution urban land cover information is essential for understanding the complex urban environment. Although publicly available land cover products offer valuable opportunities to minimize labor-intensive annotation efforts in deep learning-based mapping, effective strategies to address their inherent uncertainties remain underdeveloped. To mitigate the negative impact of label noise and enhance segmentation accuracy, we propose a Semantic-augmented and noise-tolerant Network (SantNet) for high-resolution urban land cover mapping using public products as imperfect labels. SantNet incorporates three key components: 1) a high-resolution-preserving backbone (HPB) to maintain detailed image resolution; 2) a noise-handling module (NHM) to enhance robustness against erroneous labels; and 3) a semantic augmentation module (SAM) to improve feature representation through noise-adapted contrastive learning. Benchmark comparisons with six general semantic segmentation networks and three noise-robust models demonstrate the superiority of our proposed model. Extensive ablation studies further confirm the effectiveness of each module within SantNet. Additionally, we validated SantNet's transferability across different cities, where it consistently outperformed other models, demonstrating its suitability for large-scale mapping applications. We employed this model to generate 1-m high-resolution Essential Urban Land Cover Category (EULCC) mapping for all urban areas across the Conterminous United States (CONUS). This dataset, with a validated overall accuracy (OA) of 88.77%, surpasses existing large-scale urban land cover products in spatial coverage, resolution, detail richness, and category granularity, positioning EULCC as a foundational resource for precise urban studies and applications. The data are publicly available at https://doi.org/10.5281/zenodo.13987465**

***Index Terms*—Deep learning, high resolution, land cover, noisy labels, semantic segmentation, urban area.**

I. Introduction

LAND cover, shaped by natural and anthropogenic processes, is a crucial component of the Earth system [1], [2], [3], [4]. Land cover datasets are fundamental for understanding climate change, land–atmosphere interactions, terrestrial energy balances, hydrological cycles, biodiversity conservation, ecosystem services, and planetary health [5], [6], [7], [8]. Rapid urbanization, marked by surging urban population and land cover transitions, has driven economic and social development [9], [10]. However, this expansion has also generated numerous societal, environmental, and ecological challenges [11], [12], [13]. Therefore, accurate, timely, and detailed urban land cover information is critical for mitigating these adverse effects and supporting sustainable development [14], [15], [16].

Remote sensing is one of the most powerful leverages for characterizing land cover changes due to its large-scale coverage, rapid data acquisition, easy accessibility, and rich information [17], [18], [19]. The objective of land cover classification using remote sensing imagery is to assign the most appropriate class from a predefined classification scheme to each pixel in the image. Numerous efforts have advanced classification methods for land cover classification in recent decades [18], [20], [21], with existing large-scale land cover products primarily derived from pixel- or object-based machine learning methods combining robust feature engineering and classification processes. Pixel-based methods, which exploit the rich spectral information of each pixel, have been widely used in medium- and low-resolution land

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cover mapping with acceptable accuracy [22], [23]. However, these methods often yield unsatisfactory performance in high-resolution images due to spectral confusion, limited spectral bands, and the complexity of spatial details [17], [24]. In contrast, geographic object-based image analysis (OBIA) approaches, which support multiscale analysis and incorporate additional features in spatial and textural dimensions, have been widely adopted for reducing the salt-and-pepper noise commonly seen in high-resolution mapping results [25], [26]. However, these traditional data-driven methods depend heavily on the quality of human-engineered features, necessitating extensive manual interventions and domain expertise [27], which limits their flexibility and adaptability in handling large-scale, complex land cover mapping [28]. Recently, deep learning has attracted substantial attention in the remote sensing community due to its superior accuracy and efficiency achieved in many tasks [29], [30], [31]. Unlike traditional machine learning methods, deep learning approaches integrate feature extraction and classification into a cohesive and end-to-end neural network structure that can be automatically and progressively optimized on the task-specific objectives [27], [32]. These advanced methods have not only improved traditional pixel- and object-based mapping frameworks but also facilitated the development of a new image-based paradigm—semantic segmentation—which is increasingly employed in land cover mapping.

One of the major challenges in applying deep learning over large areas is the collection of reliable labels [33]. The quantity and quality of training critically determine the accuracy and generalization capability of the developed models [34], [35]. Among existing studies, two primary methods have been used to collect accurate and sufficient training labels: interpretation- and derivation-based approaches [36]. Interpretation-based approaches depend heavily on the manual efforts of experts for visual interpretation, ensuring high data confidence in data. However, these time-consuming, labor-intensive, and costly methods greatly impede large-scale high-resolution land cover mapping applications [37] and can still incur error rates of around 10% even with the most skilled interpreters [38]. On the other hand, by individually or jointly deriving available training samples from the existing land cover products, derivation-based approaches can collect amounts of geographically distributed labels in a highly automated and cost-effective way, facilitating national or global studies [39], [40], [41]. However, labels derived this way are often subject to substantial uncertainty due to inherent product errors, scale differences, temporal mismatch, and application gaps. Regardless of the method used, label uncertainty inevitably reduces datasets quality and subsequently degrades land cover classification performance [42], [43]. In recent years, researchers have proposed various effective measures to mitigate its negative impacts on mapping over large areas [24], [33], [37], [42], [44], [45]. For example, Dong et al. [45] incorporated an online noise correction approach into the segmentation models, producing a 3-m national land cover map for China without using any human-labeled data. To reduce the adverse effects of noisy samples, Li et al. [24] developed a low-to-high network with a confident area selection module and a weak supervision strategy for high-resolution land cover mapping. Liu et al. [33] used a conditional random field model to initially refine land cover labels, followed by a class-conditional label correction method to adjust noisy labels during model training. Collectively, these studies demonstrate that explicitly addressing noise in training labels is a promising way to enhance the performance of high-resolution mapping with imperfect data. However, constrained by the classification schemes of reference land cover products, current efforts primarily target general categories such as impervious surfaces or built-up areas rather than fine-grained urban objects (e.g., buildings versus roads). A robust and generalizable noise-handling framework tailored to the complex and diverse urban scenarios has yet to be developed.

Mapping land cover in urban areas presents significant challenges because of the spatial heterogeneity and complexity of urban landscapes [10], [46]. This heterogeneity is characterized by frequent and diverse variations of objects across spatial dimensions. High-resolution remote sensing images are essential for navigating these heterogeneous landscapes and capturing detailed spatial features such as individual buildings, sidewalks, and urban green spaces [15], [47]. However, many existing semantic segmentation models compromise by reducing the original resolution to alleviate computational burden, leading to a loss of high-resolution details and smoothing effects in the results [33]. Preserving high-resolution representation throughout the feature learning is crucial for accurately delineating the spatial heterogeneity of dynamic urban environments. Additionally, the complexity of urban areas stems from the high intraclass variability and interclass similarity among urban objects, which often cause significant class confusion and complicate fine-grained urban land cover mapping [48], [49]. Tang et al. [50] developed a semantic-augmented contrastive learning module specifically aimed at enhancing semantic distinction within and across classes. Similar contrastive learning-based methods have also been used to improve semantic information in remote sensing representation learning [51]. Nevertheless, the common prerequisite of these methods is clean labels, which are critically important to guide the construction of positive and negative samples in contrastive learning, thereby limiting their broader applications in land cover mapping when working with imperfect labels.

To address the challenges outlined above, we developed a Semantic-augmented and noise-tolerant Network (SantNet) to delineate high-resolution urban land cover using aerial images and imperfect labels. This study aims to establish a practical, adaptable, and cost-effective framework for large-area, fine-resolution land cover mapping using publicly available products and advanced deep learning algorithms. In general, the main contributions of this study are summarized as follows.

1) The semantic segmentation framework SantNet integrates three complementary modules to simultaneously address the fine-scale heterogeneity of urban scenes and the presence of noisy supervision. First, a high-resolution-preserving backbone (HPB) combines a multiresolution encoder with a full-resolution branch, ensuring that detailed spatial features are retained throughout the network. Second, a noise-handling mech-

anism (NHM) employs adaptive noise filtering (NF), which isolates high-confidence pixels by thresholding the discrepancies between current predictions and uncertain labels, and consistency-based noise correction, which rectifies mislabeled pixels by evaluating class agreement across recent prediction iterations during training. These two components jointly enable stable and accurate parameter optimization by providing a cleaner and more reliable sample set. Third, a semantic augmentation module introduces noise-aware contrastive learning to strengthen feature representations and enhance class separability among complex urban land cover types.

2) We conducted comprehensive experiments to rigorously evaluate the effectiveness of the proposed framework. These include benchmark comparisons with both general-purpose semantic segmentation networks and dedicated noise-robust models, as well as detailed ablation studies to assess the individual contributions of each module and sensitivity analyses of key hyperparameters. The experimental results consistently show that SantNet outperforms existing methods, particularly under varying levels of label noise, and confirm that the proposed modules collectively drive significant improvements in segmentation accuracy.
3) Taking an open-access high-resolution urban land cover product as training data, the framework is applied to overcome label mismatches caused by land cover changes and classification uncertainties in the reference product while adapting to the diverse landscapes across regions. As a result, it generates 1-m-resolution Essential Urban Land Cover Category (EULCC) maps for 2645 urban areas across the Conterminous United States (CONUS), delineating fine-grained urban land cover at the country scale. The product has been publicly released and is intended to serve as a valuable foundational dataset for a wide range of urban studies and practical applications.

II. Methodology

A. Study Area and Datasets

This study focuses on urban areas of the CONUS, which are defined by the MAF/TIGER urban boundary [52]. In total, 2645 urban areas spanning from approximately 24.52°N to 49.38°N in latitude and from 66.95°W to 124.77°W in longitude exhibit significant variation in climatic and topographic characteristics. Such diverse natural and anthropogenic factors collectively shape the complex urban landscapes, posing considerable challenges to the nationwide land cover mapping.

The National Agriculture Imagery Program (NAIP) acquires meter- to decimeter-level aerial images with four spectral bands, i.e., blue (400–580 nm), green (500–650 nm), red (590–675 nm), and near infrared (675–850 nm), during the leaf-on growing seasons [53]. These images have been preprocessed to ensure the data quality before public release. They are promising data sources with high spatial resolution to characterize the complex urban land cover patterns. Specifically, we collected two sets of NAIP imagery for our study. The first dataset is the images captured in the same year as the label data for six selected cities, and the second dataset is the latest images until 2022 covering all CONUS urban areas.

The UrbanWatch dataset [54] is a high-quality and publicly available urban land cover dataset derived from 2014 to 2017 NAIP airborne imagery. It provides 1-m resolution, wall-to-wall land cover types for 22 major cities across the CONUS, representing a diverse range of built environments and urban landscapes. The classification scheme consists of building, road, parking lot, tree canopy, grass, agriculture, water, and others, demonstrating great potential to support various urban research topics. The reported classification accuracies for each mapped city range from 87.58% to 94.04%. To validate the effectiveness and robustness of our models, we chose six cities with high classification accuracy across different landscapes, including Chicago, Dallas, Los Angeles, Raleigh, Riverside, and Washington D.C. We regarded the land cover labels from these six cities as reliable reference labels. Subsequently, the land cover labels from all 22 cities were used to train a global model, enhancing its robustness against label noise in the reference and refining our approach to the final national 1-m urban land cover mapping.

To further verify the broad applicability of the proposed methods, we conducted additional experiments on a public dataset, the Poland dataset, which contains high-resolution imagery and multisource land cover labels across 14 provinces [55]. As exemplified in Fig. S1 (see the Supplementary Material), the dataset comprises 280 image–label pairs (1024 × 1024 pixels) with aerial imagery at 0.25–0.5 m, high-resolution ground references, and several low-resolution land cover products. We used ESA WorldCover as the noisy label source and reprojected the labels into four unified classes: built-up, tree canopy, low vegetation, and water. This dataset provides a valuable benchmark for evaluating robustness to real-world label noise. Specifically, the label noise arises from both resolution mismatches and classification errors in the products, making it more realistic and challenging than synthetic noise. Moreover, the availability of accurate high-resolution ground truth enables a reliable assessment of model performance.

B. Overall Architecture of SantNet

The proposed SantNet is an end-to-end semantic segmentation model designed to extract fine-scale land cover information from high-resolution remote sensing imagery with noisy labels. The architecture of the network is illustrated in Fig. 1. This network is built upon a HPB that consistently maintains the original image resolution throughout the hierarchical feature learning at different levels and pixel-wise classifications. We further incorporated two modules into this segmentation backbone: 1) the NHM deployed in the last layer of the network to enhance the model's robustness to the noisy labels and 2) the SAM added in the middle of the network to strengthen feature representation capability. The details of each component are given as follows.

C. High-Resolution Preserving Backbone

To improve the fine-scale semantic segmentation within high-resolution images, we employed an HPB consisting of

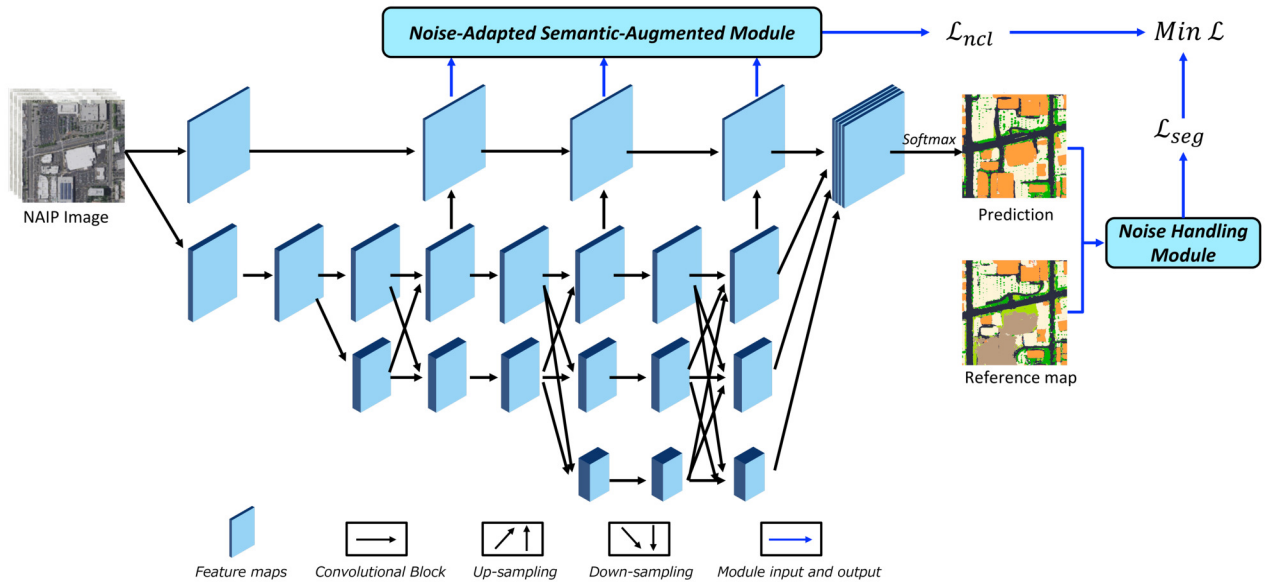


Fig. 1. Overall structure of the proposed SantNet semantic segmentation model, which is built upon an HPB with an additional full-resolution branch, a noise-adapted semantic-augmented module, and an NHM to adapt to complex landscapes and mitigate the adverse effect of significant label noise.

two branches: a multiresolution interaction branch and a high-resolution preserving branch. The multiresolution interaction branch adopts the architecture of a high-resolution network (HRNet) [56]. HRNet maintains both high-resolution and high-level representations through parallel high-to-low resolution streams and multiscale fusion, thereby enhancing spatial precision and semantic information of the feature maps for predictions. As shown in Fig. 1, this HRNet branch starts with a stem, including two stride-2 convolution (Conv)–batch normalization (BN)–rectified linear unit (ReLU) blocks, downsampling the feature resolution to 1/4, following the original network design. Using these downsampled features as input, parallel multiresolution convolution and repeated multiresolution fusion constitute the main body of the HRNet. There are four stages in total, and the number of multiresolution parallel streams for each stage increases progressively. Each stage applies residual units and cross-resolution fusion to generate high-resolution representations with strong position sensitivity and abundant semantics. At the end of the network, the multiresolution representations are rescaled to the full resolution of the input image through bilinear upsampling and concatenated together.

To compensate for the degradation of positional information caused by downsampling, we added a full-resolution branch to retain the highest resolution feature representation, which is crucial in highly heterogeneous urban environments. Besides the residual convolution units used to encode high-resolution features, fusion operations are incorporated to integrate the multiresolution semantic information from the multiresolution interaction branch. It facilitates the model learning semantically strong and spatially precise feature representation. The final full-resolution and upsampled multiresolution features are concatenated and fused via a 1×1 convolution, followed by two Conv layers and a SoftMax layer to produce pixel-wise classification probabilities.

D. Noisy Label Handling

Label noise is an inevitable challenge, especially for labels derived from existing land cover products. Neglecting label noise can cause the model to overfit on the noise, seriously undermining classification performance. Therefore, we designed the NHM to mitigate the negative impact of label noise during training. Positioned primarily at the model's output, the NHM consists of two main components: NF and NC, as illustrated in Fig. 2.

1) NF: This step aims to retain high-confidence pixels and eliminate noisy pixels for the subsequent loss calculation and model optimization. We denote the network output as $\boldsymbol{Y} = \{y_{cij} | y_{cij} \in [0, 1]\}$, where y_{cij} represents the predicted probability for class c at pixel (i, j). The provided reference set is denoted as $\bar{\boldsymbol{Y}} = \{\bar{y}_{cij} | \bar{y}_{cij} \in \{0, 1\}\}$, where $\bar{y}_{cij}$ represents one-hot encoded label. Following the idea of treating samples with smaller loss as clean samples [57], we define the confidence matrix $\boldsymbol{Q} = \{q_{ij}\}$ as the cross-entropy loss between $\boldsymbol{Y}$ and $\bar{\boldsymbol{Y}}$, calculated as follows:

$$q_{ij} = -\sum_{c=1}^{C} \bar{y}_{cij} \log y_{cij} \tag{1}$$

where C is the number of classes. Subsequently, a threshold r is set to partition each image into a confident mask $M_{\text{conf}} = \{(i, j) | q_{ij} \leq r\}$. Rather than using a fixed value, the threshold r is adaptively defined as the average of confidence values across the entire mini-batch, which is calculated as follows:

$$r = \frac{1}{BHW} \sum_{b,i,j} q_{ij}^{b} \tag{2}$$

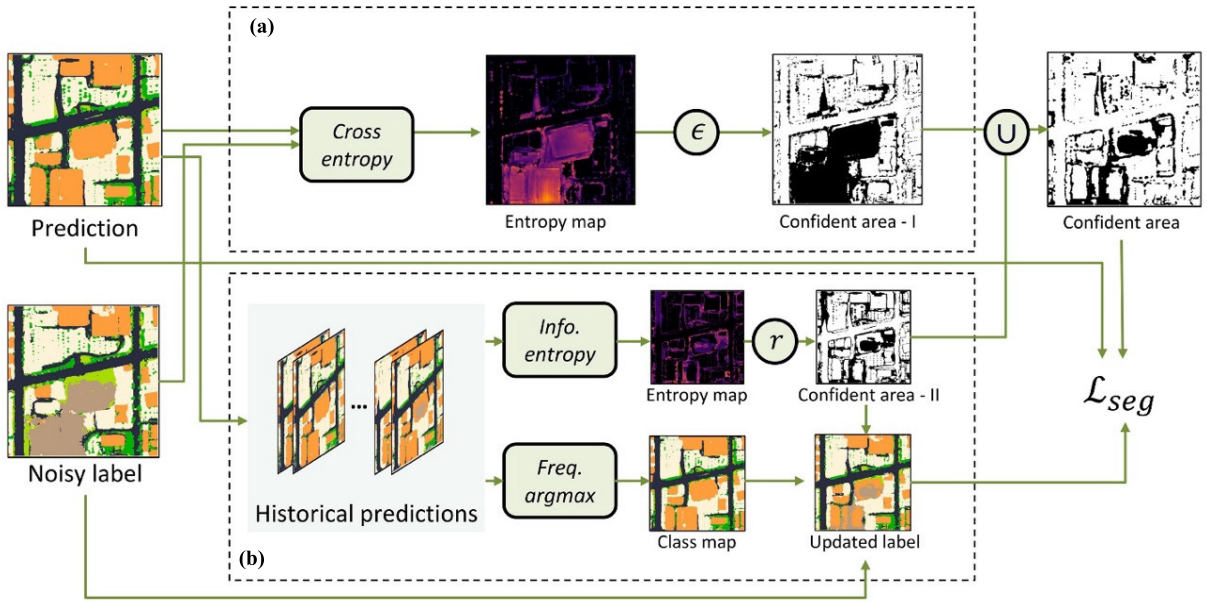


Fig. 2. Workflow of the NHM including two parallel processes. (a) NF. (b) NC.

where B denotes the batch size and H and W represent the height and width of the input image, respectively.

2) NC: The design of our NC method is motivated by key findings from previous studies, which indicate that the predictions of noisy samples are either inconsistent across different epochs or consistently match the correct class [58], [59]. Based on these observations, our framework employs a correction strategy based on the consistency of predictions to identify the candidate pixels and then update their labels. Given an image, we defined the model's output at epoch e as the class probability distribution tensor $Y^e \in (0,1)^{C\times H\times W}$, where C represents the number of land cover classes here. To convert the probability distribution into the class index matrix $\hat{Y}^e = \{\hat{y}^e_{ij}\}$, where $\hat{y}^e_{ij}$ indicates the predicted class index for the pixel located at position (i, j) in the image, we employ the argmax function to select the class with the highest probability, which can be expressed as follows:

$$\hat{Y}^e = \underset{c\in\{1,2,\ldots,C\}}{\operatorname{argmax}}\ Y^e_c \tag{3}$$

where $\hat{Y}^e \in \{1,2,\ldots,C\}^{H\times W}$ and c is the class index. To assess the consistency of the predictions, we comprehensively consider the predictions in the ongoing epoch e and the historical predictions from the previous $t-1$ epochs, forming a prediction set $H = \{\hat{Y}^{e-h} | h = 0, 1, \ldots, t-1\}$ with the condition that $\ddot{e} > t-1$. To construct H, the prediction of each image needs to be recorded in every epoch. Based on the prediction set H, we calculate the frequency of each class for each pixel in t epochs, which is defined as

$$\text{freq}^e_c = \frac{1}{t}\sum_{h=0}^{t-1} \mathbb{I}\left(\hat{y}^{e-h} = c\right) \tag{4}$$

where freq^e_c denotes the frequency of each pixel predicted as class c over the t epochs when the training is in epoch e, and $\mathbb{I}$ is the indicator function, which is 1 if the condition is true and 0 otherwise. Normalized information entropy is then adopted to quantify the uncertainty u^e o for each pixel, calculated as follows:

$$u^e = -z\sum_{c=1}^{C} \text{freq}^e_c \log \text{freq}^e_c \tag{5}$$

where the normalization factor z is equal to $1/\log C$. The pixels with lower uncertainty values indicate higher consistency in predictions across epochs, marking them as potential candidates for further refinement. When the uncertainty u^e is below a predefined threshold ϵ, the pixel is marked as the candidate for label update, and the candidate set is defined as $M_{\text{update}} = \{(i, j) | u^e_{ij} < \epsilon\}$. Then, pixels in M are updated to the class with the highest frequency f^e_c, while the rest stays the same, which is formulated as

$$\tilde{y}^e_{ij} = \begin{cases} \text{OH}\left(\underset{c}{\operatorname{argmax}}\ \text{freq}^e_c\right), & \text{if } (i,j) \in M_{\text{update}} \\ \bar{y}_{ij}, & \text{if } (i,j) \notin M_{\text{update}} \end{cases} \tag{6}$$

where $\text{OH}(\cdot)$ denotes the one-hot encoding, $\tilde{y}^e_{ij}$ denotes the updated one-hot label, and $\bar{y}_{ij}$ denotes the provided one-hot label for the pixel (i, j). Therefore, we obtain the updated labels for the current epoch, denoted as $\tilde{Y}$. Notably, this NC method performs online updates, meaning that updates occur after each minibatch rather than at the end of an entire epoch, improving computational efficiency and avoiding redundant calculations.

For each minibatch, NF and NC proceed sequentially. The outputs of the NF mask M_{conf} and NC mask M_{update} are combined to form their union set, denoted as M_{clean}, while the remaining pixels are grouped as M_{noisy}. Only the clean pixels are used to compute the cross-entropy loss between the

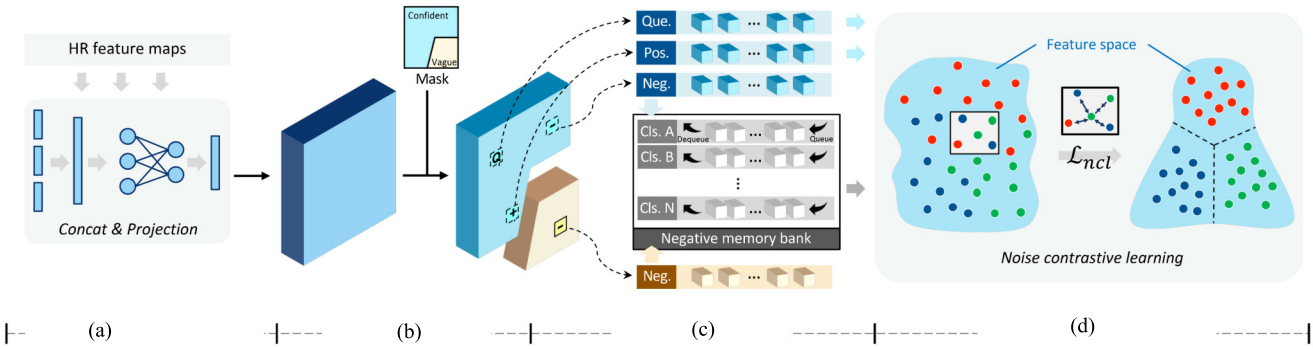


Fig. 3. Workflow of the noise-adapted semantic-augmented module (SAM) involving four stages. (a) Feature transformation, (b) spatial masking, (c) feature sampling, and (d) semantic augmentation by noise contrastive learning.

predictions $\boldsymbol{Y}$ and updated labels $\tilde{\boldsymbol{Y}}$. The segmentation loss $\mathcal{L}_{\text{seg}}$ is formulated as

$$\mathcal{L}_{\text{seg}} = -\frac{1}{|M_{\text{clean}}|} \sum_{(i,j)\in M_{\text{clean}}} \sum_{c=1}^{C} \tilde{y}_{cij} \log y_{cij} \tag{7}$$

where $\tilde{y}_{cij}$ and y_{cij} represent the updated label and predicted probability for class c at pixel (i, j), respectively. This method leverages the refined and corrected labels to guide the model's learning process, thereby reducing the potential for overfitting to the noisy data and improving the performance of land cover segmentation.

E. Noise-Adapted Contrastive Learning

Contrastive learning is a leading paradigm in representation learning, significantly propelling advances in both unsupervised and supervised learning [60], [61]. This method is characterized by the principle of pulling similar samples (positive pairs) closer together while pushing dissimilar samples (negative pairs) farther apart in the embedding space [62]. In supervised contrastive learning, the class information is leveraged to choose positive samples and negative samples for the anchor samples, constructing the positive pairs in the same class and the negative pairs in different classes. Since the method relies on the guidance of ground truths, noise may degrade the whole representation learning process by wrongly pulling and pushing samples. The most common strategy for handling noisy pixels in contrastive learning is discarding the unreliable pixels, which causes information loss during training because some pixels may never contribute to learning. By jointly using high-confidence labeled samples, potential semantic information from pixels with unknown or inaccurate labels can be exploited at the feature level, improving model performance [24], [63]. Moreover, even when a model trained with noisy labels may become confused between a few classes for a particular pixel, the output probability distribution often provides useful information by indicating which classes the pixel is least likely to belong to [64]. Based on this characteristic, such unreliable pixels can be convincingly treated as negative samples to those most unlikely classes, enabling the model to leverage them for learning the semantic relationship between pixels. Building on this insight, we proposed a novel module named SAM, which uses noise-adapted contrastive learning to enhance the representation even in the scenario of noisy labels.

The workflow of our noise-adapted SAM is illustrated in Fig. 3. It operates on the high-resolution feature representation $\boldsymbol{\mathcal{F}}$ of intermediate layers. These feature maps are concatenated and passed through a projector comprised of two 1×1 convolutional layers followed by normalization, producing embeddings used for the following computation of contrastive loss. The key step in SAM is constructing the query pixels, corresponding positive samples, and negative samples. M_{clean} and M_{noisy} generated in the NHM are reused in this method to differentiate the reliable and unreliable pixels. We randomly sample pixels for the classes present in the current batch as query pixels. The feature set of these pixels for class c is denoted as

$$\boldsymbol{Q_c} = \left\{ f_{ij} \middle| \tilde{y}_{cij} = 1, (i,j) \in M_{\text{clean}} \right\} \tag{8}$$

where f_{ij} is the representation of the pixel (i, j) in $\boldsymbol{\mathcal{F}}$. Normally, the candidate positive samples are those pixels with the same class label as the query pixel. To reduce the computational consumption, we calculate the feature center of pixels in each class as the positive sample

$$f_c^+ = \frac{1}{|\boldsymbol{\mathcal{F}}_c|} \sum_{f_{ij}\in \boldsymbol{F}_c} f_{ij} \tag{9}$$

where $\boldsymbol{\mathcal{F}}_c$ represents the feature set of pixels belonging to class c. This means that only one class-specific positive feature vector is used for query samples from the same class in each mini-batch. For constructing the negative samples, two schemes are used to collect different kinds of negative samples based on whether they are reliable pixels or not. For the reliable pixels, the candidate negative sample for class c should satisfy the following requirements: the sample's updated label does not belong to class c, but it is easy to get confused by class c when doing prediction. It can, therefore, be defined as follows:

$$\boldsymbol{N}_c^r = \left\{ f_{ij} \middle| \tilde{y}_{cij} \neq 1, 0 \leq \text{argsort}_c y_{ij} < r_l, (i,j) \in M_{\text{clean}} \right\} \tag{10}$$

where the argsort function is used to obtain the ascending rank of probability for class c and r_l is the low-rank threshold.

For the unreliable pixels, the requirement for qualified negative samples is that the sample likely does not belong to class c. The candidate negative samples from unreliable pixels can be defined as follows:

$$\boldsymbol{N}_c^u = \left\{ f_{ij} \middle| \tilde{y}_{cij} \neq 1, \text{argsort}_c y_{ij} > r_l, (i,j) \in M_{\text{noisy}} \right\}. \quad (11)$$

Since negative sample diversity is critical for contrastive learning [61], we employ a memory bank strategy to store the features of negative samples from previous minibatch computations. The memory bank consists of C class-wise queues with a fixed capacity. During training of each mini-batch, a certain number of features sampled from $\boldsymbol{N}_c^r$ and $\boldsymbol{N}_c^u$ are inserted into the corresponding queue according to their class, and the oldest samples are discarded when the capacity is reached.

Finally, we used the contrastive loss to realize the idea of pulling positive pairs close and pushing negative pairs apart. For efficient computation, we sampled n_{query} query pixels from $\boldsymbol{Q_c}$ to form feature subsets $\boldsymbol{F}_c^Q$. n_{neg} negative samples are sampled from the current memory bank to form the negative feature subsets $\boldsymbol{F}_c^N$. The noise contrastive learning loss $\mathcal{L}_{\text{ncl}}$ in our study is measured by the InfoNCE [65], which is formulated as

$$\mathcal{L}_{\text{ncl}} = -\frac{1}{C' \times n_{\text{query}}} \times \sum_{c=1}^{C} \sum_{f_c \in \boldsymbol{F}_c^Q} \log \frac{e^{\text{sim}\left(f_c \cdot f_c^+\right)}}{e^{\text{sim}\left(f_c \cdot f_c^+\right)} + \sum_{f_c^- \in \boldsymbol{F}_c^N} e^{\text{sim}\left(f_c \cdot f_c^-\right)}} \quad (12)$$

where C is the total number of classes, C' is the number of classes appeared in the current mini-batch, and $\text{sim}(\cdot,\cdot)$ means the cosine similarity of two feature vectors. The final optimization target of the whole network is to minimize the overall loss $\mathcal{L}$

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda \mathcal{L}_{\text{ncl}} \quad (13)$$

where λ denotes the weight of contrastive loss. Since the contrastive learning module is an auxiliary branch, its weight is usually set to a smaller value to ensure that the optimization process is primarily directed toward the segmentation task.

III. Experimental Design

A. Experiment Setting

In this study, we conducted three groups of experiments, including method evaluation, ablation studies, and country-scale mapping practices, all using the datasets mentioned in Section II-A. For the method evaluation, we benchmarked against state-of-the-art semantic segmentation models and noise-robust segmentation methods. The ablation study was designed to conduct a thorough investigation of the modules and hyperparameters in the proposed SantNet. The benchmark comparison experiments were mainly executed on the high-accuracy urban land cover dataset from six selected cities where synthetic label noise was artificially introduced, while the ablation study was implemented on the Washington D.C. dataset. The last experiment, country-scale urban land cover mapping, involved training the noise-robust model using the entire UrbanWatch land cover product as labels and performing inference across all urban areas of CONUS.

B. Noise Simulation

To quantify the accuracy and robustness of our methods, we regarded the dataset containing highly accurate land cover maps for the selected six cities as clean labels and then artificially corrupted these labels with various noise rates and modes. We specifically implemented two types of noise: pixel-level noise and object-level noise, to evaluate the model's robustness to both small-scale localized errors and broader homogeneous misclassification.

For pixel-level noise, a subset of pixels was randomly selected as candidates for corruption. Rather than changing ground truth labels to any other classes with the same probability, we biased the label changes toward easily confused classes. This approach generates more realistic noise while increasing classification difficulty. To achieve this, we first trained a simple classifier and used it to obtain the predicted class probabilities of pixels and then sampled a new class based on this distribution, excluding the true class to serve as the corrupted label.

For object-level noise, a simple linear iterative cluster (SLIC) algorithm was used to segment the image into superpixels as object-based analysis units for candidate selection and asymmetric label transformation. Based on spatial proximity and color similarity, SLIC clusters pixels into compact and approximately uniform superpixels. Through empirical parameter tuning, the average size of the generated superpixels was approximately 660 m^2, chosen to balance homogeneity within each region and avoid excessive fragmentation of land cover features. The ground truth label of each object is determined by the dominant class of pixels inside, while the class probability distribution for each object was derived from the average distribution of these pixels.

For each image in the training set, we randomly corrupted 0%–50% of pixels and objects simultaneously to simulate varying and challenging scenes of noisy labels, while the test set remained unchanged for performance evaluation. In the end, the simulated noise rates in training sets of the six cities ranged from 30% to 40%.

C. Evaluation Metrics

Following the evaluation protocols applied in land cover mapping studies, we used class-wise $F1$-score ($F1$) and overall accuracy (OA) to assess the performance of classification methods. We also applied the confusion matrix involving user accuracy (UA) and producer accuracy (PA) to evaluate the generated products. In addition to the aforementioned metrics for measuring the consistency between predictions and ground truths, we further employed the pixel-wise discrimination distance (PDD) to evaluate the intercategory and intracategory affinities of feature representations [66]. It is computed as

$$\text{PDD}_c = \frac{1}{|F_c|} \sum_{f \in F_c} \frac{\text{sim}\left(f, u^c\right)}{\sum_{i=1, i \neq c}^{C} \text{sim}\left(f, u^i\right)} \quad (14)$$

where PDD_c denotes the PDD values of class c, u^c denotes the local centroid of the pixels belonging to class c, and $|F_c|$ is the cardinality of feature set F_c.

D. Implementation Details

All the experiments were implemented using the PyTorch framework and Python on a workstation with an NVIDIA RTX 4090 GPU. The aerial images and the corresponding land cover maps were first cropped uniformly into tiles of 512×512 pixels. For each city, we selected 80% of the tiles for noise injection and training, while the rest were left for testing.

The Adam optimizer was used to optimize the parameters of SantNet with a batch size of 4. For the hyperparameters in SantNet, we adopted uniform settings below to ensure consistent and fair benchmark comparisons and ablation studies, unless otherwise specified. In the process of noise handling, the number of revisit epochs t was set to 5, and the uncertainty threshold ϵ was set to 0.05. For the noise-adapted contrastive learning, the numbers of pixel samples n_{query} and n_{neg} were set as 200 and 1024 for each class, respectively, and the length of the class-wise queue in the memory bank was 60 000, balancing the computational efficiency and sample diversity. The weight λ of contrastive loss in calculating L was set as 0.001.

Training followed a two-stage strategy: warm-up training and formal training. During warm-up, SantNet was trained from scratch without noise handling or contrastive learning to allow the network to acquire sufficient feature discrimination for subsequent noise differentiation and correction. After warm-up, the NHM and SAM modules were activated to enable noise-robust training.

Given the different sample sizes across six cities, two sets of parameters related to the training epoch were designed. For the Raleigh and Washington D.C. datasets with relatively small sizes, we trained 20 epochs in warm-up stages and 80 epochs in official training stages. For the rest of the cities, the numbers of epochs trained in two stages are 10 and 40. The initial learning rate was 0.01, decayed by a factor of 0.1 using a scheduler, with an interval of ten epochs for small cities and 20 epochs for others.

For CONUS-wide mapping, we first matched the NAIP images with the land cover map of 22 cities from the UrbanWatch dataset, both of which were further cropped into image–label pairs of 512×512 pixels. Using this dataset, we trained the proposed models for a total of 50 epochs, with the first ten epochs for warm-up training and the remaining epochs for formal training with active noise processing. The values of other hyperparameters were kept the same as previously mentioned. During the inference stage, we consistently cropped the NAIP imagery covering the urban area of CONUS into tiles. We then used the well-trained SantNet to predict the land cover category for these tiles, which were subsequently stitched together to produce the 1-m EULCC dataset for CONUS.

IV. Results

A. Benchmark Comparisons

In order to validate the superiority of the proposed method, we selected ten advanced semantic segmentation methods for comparative experiments, including general segmentation models and tailored noise-robust models. These general segmentation methods included both CNN-based models (i.e., DeeplabV3+ [67], A2FPN [68], and MANet [28]) and Transformer-based models (i.e., Segformer [69], UnetFormer [70], and DCSwin [71]), all of which have been demonstrated to perform well in land cover mapping from remote sensing imagery.

Regarding the noise-robust methods, the first, denoted as Model-S hereafter, relies on the symmetric cross-entropy (SCE) loss, which addresses insufficient training and overfitting problems on noisy datasets [72]. The second method, denoted as Model-C, is derived from the framework proposed by Cao and Huang [73], which uses a noisy label correction approach for thematic mapping. The third method, denoted as Model-D from Dong et al.'s [45] work, also employs an online correction protocol, yet with distinct ways of uncertainty measurements and thresholding for multiclass land cover mapping. Paraformer is an efficient weakly supervised framework leveraging pseudo-label-assisted training to bridge the resolution gap for high-resolution land cover mapping [55]. Eleven models, including SantNet and the aforementioned methods, were trained in the same corrupted training set. We evaluated the performance of these methods on pure test sets of the selected six cities.

The quantitative results of benchmark comparisons are summarized in Table I. Across all six cities, the proposed SantNet consistently achieved the best performance in OA, significantly outperforming the other benchmark methods, as shown in Fig. S2 (see the Supplementary Material). Among the general segmentation models, MANet performed best for Chicago, Los Angeles, and Riverside, while UNetFormer was optimal for Dallas, Raleigh, and Washington D.C. Even though there is no explicit treatment for noisy labels, they still achieved competitive classification performance comparable to some of the noise-robust methods because of their advanced network architectures.

Except for the proposed SantNet, the four noise-robust models performed differently across the six cities, with Model-D achieving the best results in Chicago, Raleigh, and Washington D.C., Model-S in Los Angeles and Riverside, and Model-C in Dallas. Paraformer did not achieve clear superiority in any of six cities, yielding suboptimal performance relative to the other models. Our methods achieved a superior performance with OA 2.20%–3.62% higher than the best-performing noise-robust method across the six city datasets. This indicates that our introduced modules surpass benchmark methods in mitigating the negative impact of noisy labels on land cover classification. More details on the class-specific metrics are provided in Table SI (see the Supplementary Material).

The proposed methods consistently achieved superior $F1$-score in major classes (building, tree, grass, parking, and road) as well as the relatively easy class (water), while performance in minor classes (barren, agricultural, and others) varied significantly across cities. In addition to the labels directly from UrbanWatch, we also carried out additional manual labeling for heterogeneous scenes across six cities. A total of 150 tiles, annotated pixel-by-pixel by domain experts, were collected to evaluate model performance more precisely. Using this manually curated dataset, we recalculated the accuracy and confirmed the superiority of the SantNet, with only minor

TABLE I

QUANTITATIVE COMPARISON OF CLASSIFICATION PERFORMANCE AMONG BENCHMARK METHODS ACROSS SIX CITIES IN TERMS OF OA USING URBANWATCH LABELS AND MANUAL ANNOTATIONS

Model	Chicago		Dallas		Los Angeles		Raleigh		Riverside		Washington D.C.	
	OA(U)	OA(M)	OA(U)	OA(M)	OA(U)	OA(M)	OA(U)	OA(M)	OA(U)	OA(M)	OA(U)	OA(M)
DeeplabV3+	81.36	84.36	86.26	84.06	81.78	78.43	87.88	86.47	75.78	68.83	79.08	80.73
A2FPN	81.37	84.15	85.82	85.97	81.04	75.19	88.90	86.81	78.49	74.15	80.22	80.9
MANet	82.66	85.36	86.69	86.07	83.48	80.27	89.68	87.04	80.82	75.49	81.85	82.81
Segformer	79.80	81.23	82.25	84.46	81.20	77.67	88.85	86.15	76.02	69.98	78.01	80.53
UNetFormer	82.21	84.36	87.36	84.54	82.94	79.54	90.19	87.86	79.16	73.78	81.96	83.67
DCSwin	79.89	82.82	83.80	78.84	76.44	71.75	86.03	83.37	76.22	70.43	79.60	80.04
Model-S	82.20	83.78	85.29	82.8	83.72	81.1	86.05	85.8	82.21	73.68	82.28	82.14
Model-C	79.01	80.03	85.87	86.22	78.07	68.69	87.35	86.49	67.06	71.04	81.02	81.59
Model-D	82.94	85.10	83.23	78.51	74.85	74.12	89.48	88.21	78.92	72.68	84.82	86.34
Paraformer	75.86	75.12	83.43	79.18	78.10	72.25	85.18	79.17	74.59	68.58	81.30	83.08
SantNet	86.01	87.88	89.22	90.31	86.90	84.65	92.37	90.53	84.41	78.01	88.44	89.19

OA (U) refers to the OA calculated using UrbanWatch labels, while OA (M) refers to the OA calculated using the manually annotated dataset.



Fig. 4. Visual comparison of land cover mapping results on six scenes derived from benchmark methods.

differences compared to the results calculated using the full UrbanWatch labels. This consistency not only reaffirms the advantage of SantNet but also supports the validity of the original dataset for evaluation.

To intuitively illustrate the differences among comparative methods for visual inspection, six representative scenes with diverse landscapes across cities are shown in Fig. 4. Compared with two general segmentation models and three noise-robust methods, the proposed SantNet produced results exhibit higher consistency with the actual land cover distribution shown in six scenes.

In the first scene, high-density urban area with buildings of varying sizes and scattered green space, methods performed well in identifying most of the urban objects. However, differences arose in classifying the large buildings and their surroundings in the scene center. All methods except SantNet

and Paraformer misclassified parts of the large building on the right as parking, while Paraformer misclassified those parking lots as road and building. Additionally, the two general segmentation methods failed to accurately identify the parking area on the left.

The second scene, a suburban area dominated by agricultural land, reveals that UNetFormer struggled to maintain the integrity of the large continuous land cover while MANet significantly underestimated small-sized and sparsely distributed trees. Paraformer failed to correctly identify most of the agricultural land, misclassifying it as grassland in this scene. The third scene is a relatively challenging scene with three types of impervious showing high spectral similarity, leading to serious confusion for four comparative models. In contrast, SantNet demonstrates a stronger ability to distinguish these confused pixels in the same place, probably due to the benefits of noise-adaptive contrastive learning that enhances feature separability.

In the fourth scene, our method demonstrates significant advantages in recognizing extensive barren land with higher integrity and internal class consistency than the other methods. In the fifth scene, the proposed method also obtains better performance in capturing building boundaries. In the final scene, the main differences among methods lie in distinguishing the "Others" category. Our method and Model-D produced maps closest to the ground truth for this category. The two general models continued to struggle in differentiating between the impervious classes.

Fig. S3 (see the Supplementary Material) illustrates three representative scenarios in which all comparative models show varying degrees of underperformance. The first scenario comprises dense built-up areas, including both high-rise high-density and low-rise high-density urban street blocks. In high-rise clusters, strong shadows often obscure ground-level features such as vegetation and impervious surfaces. In low-rise dense settings, buildings and surrounding built-up elements often lack clear boundaries, complicating the separation of individual structures.

The second scenario involves sparse, multiscale low-rise built-up areas, where large commercial or industrial buildings coexist with open spaces. Here, the rooftops of large buildings and adjacent parking lots share similar textures and spatial patterns in aerial imagery, leading to inaccurate delineation of building boundary and substantial confusion between these two classes.

The third scenario features mixed open land, particularly construction sites or bare ground. Such environments are particularly challenging due to their semantic ambiguity, often leading to frequent interclass confusion among barren land, parking lots, road, and water, and consequently a notable drop in classification accuracy.

Tables SII and SIII (see the Supplementary Material) present the confusion matrices of SantNet and the baseline HRNet model, respectively, evaluated on the test sets of all six cities under the simulated noise setting. Across all classes, SantNet achieves higher PA values, with the most notable gain (+30.43%) in the "Others" category, which also records the largest improvement in UA. For UA, all categories improved compared to HRNet, except *Building*, which shows a marginal decrease of 1.02%. In terms of correctly classified sample counts, every class demonstrates increased accuracy.

For commonly confused impervious-surface classes (*Building*, *Parking*, and *Road*), improvements largely stem from reduced mutual misclassification. *Tree* and *Grass*, another pair of easily confused pair, benefit from enhanced class separately in SantNet. *Water*, being a distinct and easily recognized class, maintains consistently high accuracy in both models, with only minor UA and PA differences. In the *Agriculture* class, the PA gain is primarily due to confusion with *Grass*, while the UA increase in *Barren* results from fewer misclassifications with impervious surfaces and vegetation. These findings confirm that SantNet not only enhances OA but also substantially mitigates interclass confusion, particularly for challenging urban land cover categories.

We further benchmarked SantNet against three strong comparative methods on the Poland dataset, including UNetFormer, MANet, and Model-D, which performed well on previous evaluations. As shown in Table SIV (see the Supplementary Material), all methods achieve lower accuracies compared to the simulated noise setting, as the larger resolution gap injects substantial label noise, making the tasks more challenging. Despite this, SantNet attains the highest OA and achieves the top $F1$-score in all classes, with the largest gain (8.97%) in the *Water* class. These results on a genuinely noisy benchmark dataset further demonstrate the robustness and effectiveness of our model for land cover mapping when only imperfect labels are available.

B. Ablation Studies

The proposed SantNet incorporates several technical advancements, including HPB, NHM, and SAM, to improve its performance in high-resolution land cover mapping in the presence of noisy labels. To assess their individual and combined contributions, we conducted comprehensive ablation experiments on various SantNet variants. To examine the effectiveness and compatibilities of the proposed components, we trained and evaluated eight models, including SantNet and seven other variants. The comparative results are summarized in Table II. Note that we separately explored NF and NC of NHM in this ablation study. As summarized in Table II, HPB and NC independently improved OA by 2.49% and 1.06%, respectively, compared with the baseline model (SantNet without any of these components). While the baseline with NF alone achieved lower OA than the baseline, it still yielded higher $F1$-scores for most classes. When these components were combined, we observed strong synergy, with notable gains in both overall and class-specific metrics—indicating good compatibility among modules. Among them, the SantNet integrating all four components outperformed all variants, obtaining the highest values of OA and $F1$-scores for each class except *Water*. Removing any single noise-handling component led to a performance drop, indicating that these modules play a complementary role in SantNet's superior performance. The most significant decline is observed in the variant missing the NC step, followed by NF and then SAM. Although SAM contributed less to the overall metrics, the

TABLE II
QUANTITATIVE COMPARISON OF CLASSIFICATION PERFORMANCE AMONG MODELS WITH DIFFERENT CONFIGURATIONS

Components					F1-score							
HPB	NF	NC	SAM	OA	*Building*	*Tree*	*Grass*	*Parking*	*Road*	*Water*	*Barren*	*Other*
				82.03	83.61	89.66	64.29	59.32	77.49	91.83	20.70	65.99
✓				84.52	83.52	91.93	70.22	71.78	80.53	88.35	22.46	69.30
	✓			81.88	84.29	88.89	67.76	68.45	77.80	92.49	0	70.01
		✓		83.09	83.27	91.18	70.53	57.74	77.89	91.83	0	66.95
✓		✓	✓	85.54	85.96	92.72	71.98	65.79	81.89	91.26	0	69.26
✓	✓		✓	84.98	83.82	93.08	72.94	68.41	80.35	87.74	0	68.85
✓	✓	✓		87.68	87.85	93.60	76.97	75.89	84.66	92.83	21.04	75.52
✓	✓	✓	✓	88.44	88.46	93.98	78.47	79.83	85.48	92.41	41.12	77.74

A checkmark "✓" denotes the inclusion of the corresponding module in the model for each ablation experiment; The F1-score of 0 indicates that the model classified zero pixels as the specific class.

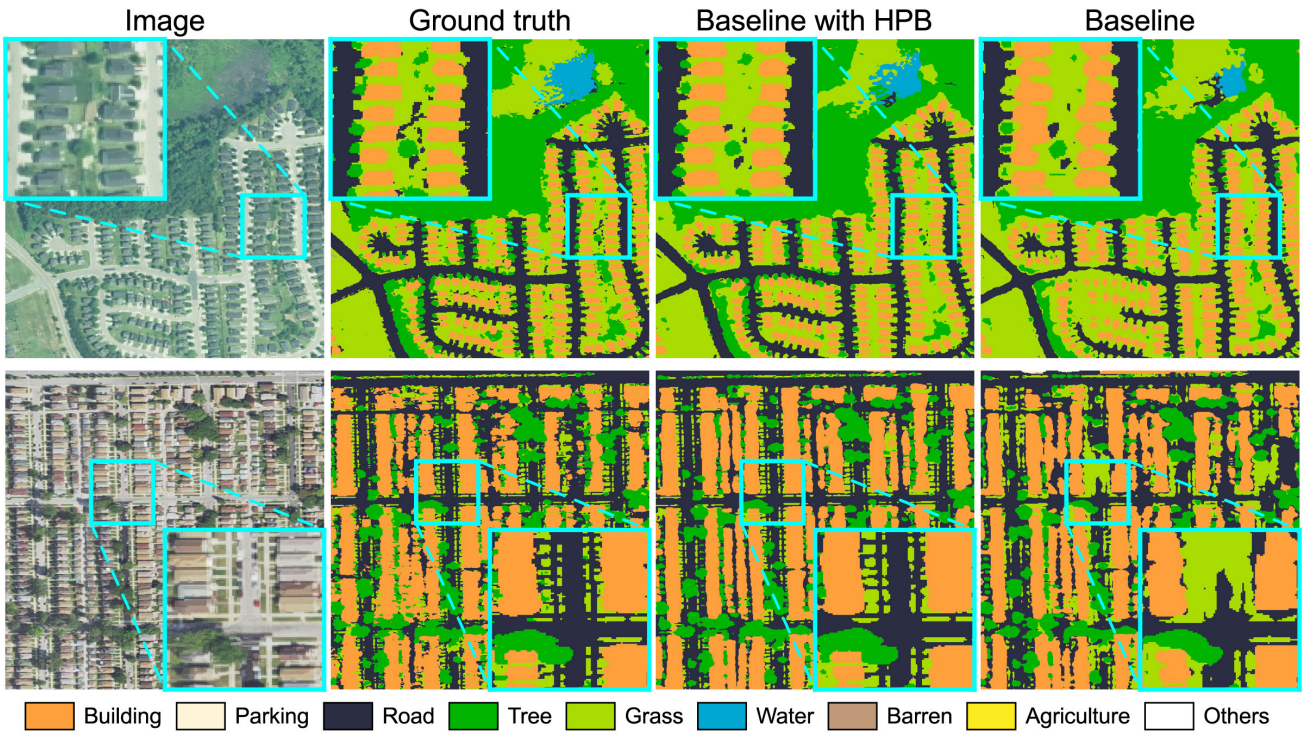


Fig. 5. Visualization of land cover mapping results by baseline model and variant with the HPB where the blue frame indicates a zoomed-in view of the selected region.

results clearly demonstrate that it helped SantNet substantially improve the classification of Barren land, a minor class in the Washington D.C. dataset. Furthermore, we observe that among all models, only those employing either NF or NC performed subpar in the minor class, underscoring the importance of integrating both steps.

To further verify the effectiveness of the proposed modules, more in-depth comparisons and analyses are given here. We selected two scenes to compare the results of the baseline model and baseline with the HPB in detail. As shown in Fig. 5, the high-resolution branch significantly improved the model performance, particularly for small objects, elongated structures, and fine boundaries. In the first scene, road connectivity was better preserved. The zoomed-in view revealed that the improved model can better distinguish adjacent building instances, whereas the baseline model performed poorly in detecting the gaps between buildings, leading to more adhesion of adjacent buildings. In the second scene, a zoomed-in view demonstrated that the HPB-enhanced model effectively recognizes subtle grassland features in a complex context, achieving better consistency with the ground truth.

To better illustrate the mechanism and the effect of the NC module, Fig. 6 shows the accuracy dynamic on the test set throughout the training process for two models: the complete SantNet and the SantNet without NHM. In the early stage, both models performed similar trends—accuracy rose rapidly at first and then plateaued as training approached the 20th epoch. After the NHM was activated at the 20th epoch, the accuracy of SantNet increased rapidly again, while the accuracy of the degraded model fluctuated within a certain range until the 30th epoch. In contrast, the NHM-free model's accuracy fluctuated within a narrow range until the 30th epoch. The learning rate

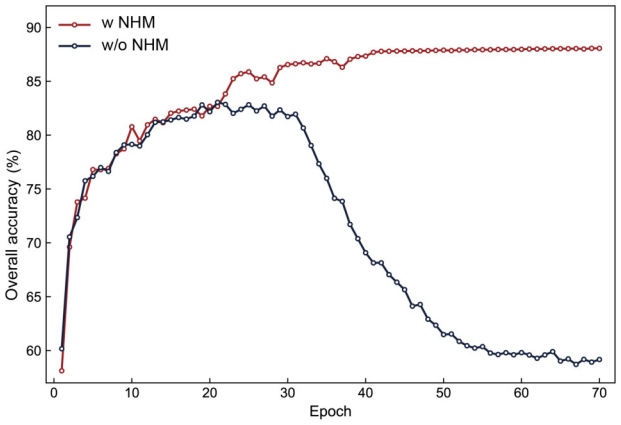


Fig. 6. OA curves showing the variation of SantNet and its variant without the NHM during training.

scheduler started to adjust the learning rate after the 30th epoch, normally allowing the model to fine-tune its parameters and converge to a more optimal set of parameters. In this training stage, the model without NHM declined continuously and sharply. This can be attributed to the model shifting its focus to learning patterns from more complex and difficult samples after the learning rate was reduced. However, for the model without noise-handling mechanisms, it could not effectively differentiate between genuinely difficult samples and mislabeled samples. As a result, it treated mislabeled samples as informative but challenging data, leading to overfitting on noisy labels and ultimately causing optimization in the wrong direction. In contrast, SantNet maintained a steady accuracy gain due to the explicit filtering and correction of noisy labels, enabling it to utilize high-confidence and more accurate labels for parameter optimization.

To further explore the robustness to varying noise levels, we evaluated the model performance across different noisy label settings, of noisy labels including the correction rate on training and the prediction accuracy on the test sets. As illustrated in Fig. 7, the overall trend in accuracy for both datasets is consistent; the indices dropped as the noise rate increased, with a sharp decline once the noise exceeded 50%.

Besides, when the noise ratio is below 50%, deep learning models exhibit greater robustness to pixel-level noise than object-level noise. For object-level noise, the model with the NHM module consistently improves the accuracy, with such improvement becoming more remarkable as the noise rate increases up to 50%.

For pixel-level noise, the NHM module does not significantly improve performance until the noise rate exceeds 30%. When the noise rate is between 30% and 50%, the NHM enhances the model's ability against pixel-level noise. For both types of noise, when noisy labels accounted for half or more of the entire label set, the improvement brought by the NHM module was marginal.

Regarding object-level noise, we additionally employed the segment anything model (SegAM)—a recent foundation model for image segmentation known for its strong zero-shot generalization ability in delineating objects without additional training. We compared the impact of using SegAM-based masks with that of the original SLIC superpixels in simulating object-level label noise across varying noise rates. The results are presented in Fig. S4. Overall, the NHM module consistently improves the performance under both SLIC- and SegAM-based noise across most noise levels. For the training-set correction rate, the performance gain under SegAM-based noise is larger than that under SLIC-based noise at all noise levels except in the 40%–50% noise range. For the test accuracy, the gain achieved under SegAM-based noise is greater than that under SLIC-based noise at all noise rates, except when the noise rate falls between 40% and 50%. Up to 40% noise rate, the improvements introduced by NHM are comparable under both noise types. Between 40% and 50%, SantNet performs slightly better under SLIC-based noise. However, when the noise rate exceeds 50%, SantNet continues to offer measurable improvement in SegAM-based noise, while its effectiveness under SLIC-based noise becomes marginal.

Finally, we quantitatively evaluated the effectiveness of the SAM by measuring the PDD index. Fig. 8 shows that the features from SantNet achieve higher PDD values than the model without SAM, suggesting that SAM successfully improves the feature space where similar pixels are closer and dissimilar pixels are farther apart, thereby optimizing the feature representation through intermediate supervision and ultimately improving the classification accuracy.

C. Hyperparameter Analysis

The proposed SantNet introduces hyperparameters that affect land cover mapping performance, including the number of revisit epochs (t), the uncertainty threshold (ϵ), and the weight of contrastive loss (λ). To test the model's sensitivity to these hyperparameters, we conducted a series of comparative experiments with varied values for each parameter. Since t and ϵ are important factors in the step of NC, we simultaneously investigated their impact on the classification performance, as shown in Fig. 9. For OA [Fig. 9(a)], the highest value is achieved by the setting ($t = 5$ and $\epsilon = 0.1$). We also found that when one hyperparameter increases or decreases, appropriately adjusting the other hyperparameter helps maintain high OA. Similar phenomena are observed in the other two subfigures. However, unlike OA, higher mean $F1$—averaged over all classes [Fig. 9(b)]—is often achieved with the combination of larger t and ϵ. This can be largely attributed to the fact that the $F1$-score for minor classes is improved using larger hyperparameters, as illustrated in Fig. 9(c).

Although SAM has been proven effective for performance improvement, its effectiveness largely depends on the setting of the loss weight (λ), as shown in Table III. Among the eight experiments, setting $\lambda = 0.001$ achieved the best overall performance. Within the range of 0.0001–0.1, the model's performance remains relatively stable. However, when λ is further increased, both overall metrics and class-wise $F1$-scores decline, and for some classes, the accuracy even drops to zero. This is mainly because larger λ gives undue greater priority to the contrastive loss, which is intended to be an

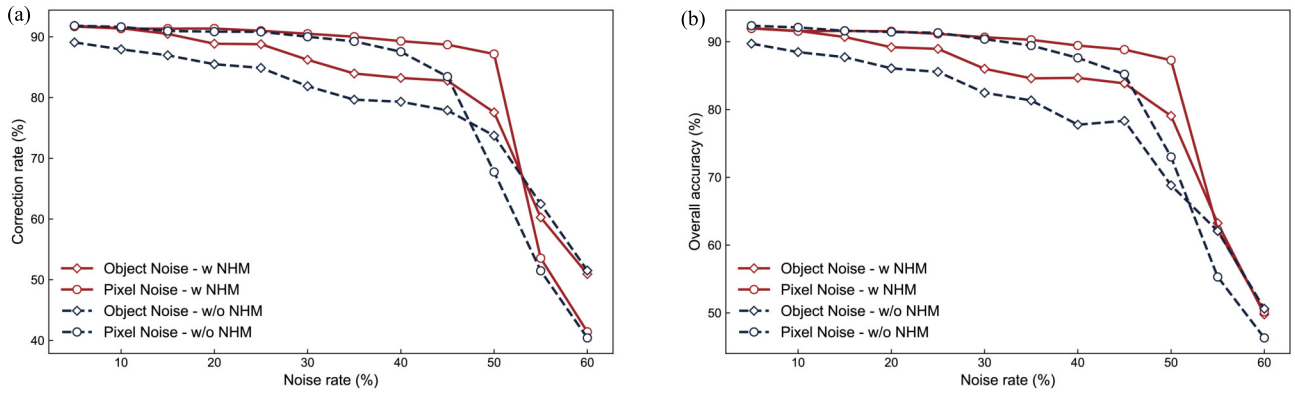


Fig. 7. Performance variations with different rates of label noise in terms of (a) correction rate in training set and (b) OA in test set.

TABLE III
PERFORMANCE SENSITIVITY TO THE WEIGHT OF CONTRASTIVE LEARNING LOSS IN TERMS OF OA AND CLASS-WISE *F*1-SCORE

Value	F1-score								
	OA	*Building*	*Tree*	*Grass*	*Parking*	*Road*	*Water*	*Barren*	*Other*
0.0001	88.19	88.15	93.89	78.11	78.97	85.04	91.09	38.16	82.56
0.001	88.44	88.46	93.98	78.47	79.83	85.48	92.41	41.12	77.74
0.01	88.28	88.18	93.93	78.58	78.86	85.22	92.67	36.21	80.18
0.1	88.09	87.95	93.77	77.87	78.79	85.14	92.42	66.70	78.07
0.5	87.10	87.62	93.41	76.19	77.21	84.32	89.32	9.38	0
1	85.07	85.52	92.75	73.50	63.02	81.76	85.58	0	0
5	79.55	81.46	91.17	67.99	0	74.13	0	0	0
10	59.38	0	85.40	8.70	0	55.31	0	0	0

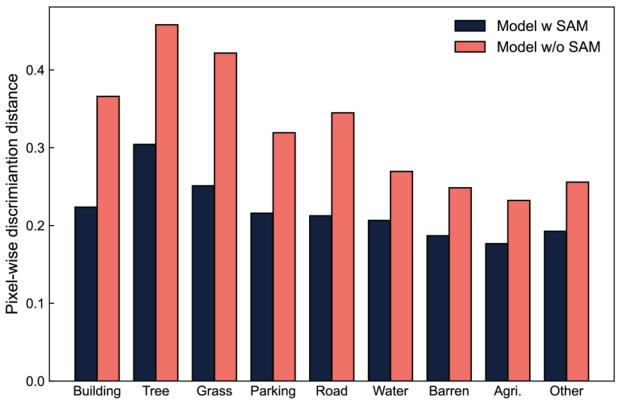


Fig. 8. Quantitative comparison of the discrimination of intermediate features generated in SantNet and the variant without semantic-augmented module (SAM).

auxiliary task. As a result, the training process overlooks the real objective function that is directly related to the classification outputs.

D. Model Transferability

To evaluate the model's generalization capability across untested urban environments, we conducted a transferability experiment. Unlike the original setting—where training and testing were conducted within the same city—this experiment used one city as the test city and the remaining five cities as the training set. This setting allowed us to assess how well the model performs when exposed to new urban landscapes.

The accuracy results for both transfer and local models across the six cities are summarized in Fig. 10. SantNet consistently outperformed the benchmark models in both local and transfer settings across all six cities, demonstrating strong transferability and highlighting its potential for large-scale land cover mapping. However, all transfer models showed a slight performance decrease compared to their local counterparts. This can be attributed to the unique urban landscapes of each city, which likely caused a performance decline. Among the test cities, Washington D.C. experienced the smallest drop in accuracy, while Los Angeles showed the largest decline. This suggests that training data from the other cities did not adequately capture the distinctive characteristics of Los Angeles, leading to a larger performance gap. These findings emphasize the importance of using a diverse and broad training dataset that covers various urban environments to ensure better transferability and generalization.

E. Comparison With Foundational Models

In addition to general semantic segmentation models mentioned above, we also evaluated two latest remote sensing foundation models—geospatial foundation model (GFM) [74] and SatLas [75]—to examine whether large-scale pretraining alone imparts inherent robustness to label noise. Both models were fine-tuned on the Washington dataset under two settings: 1) *Clean*, where the training data contained no simulated label noise, and 2) *Noisy*, where the training label was corrupted.

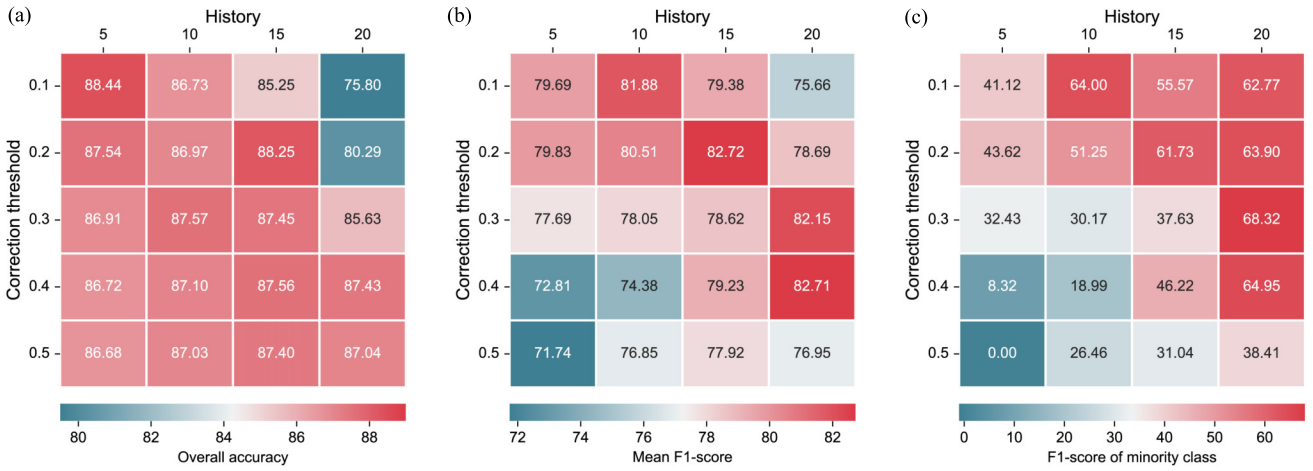


Fig. 9. Sensitivity of model performance including (a) OA, (b) mean $F1$-score, and (c) $F1$-score of minority class, to the number of revisiting epochs and correction threshold.

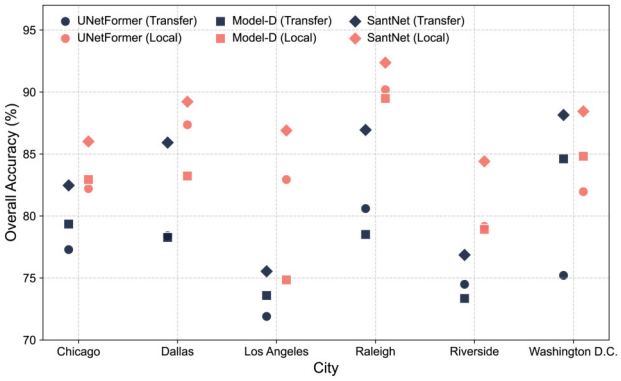


Fig. 10. Comparative evaluation of cross-city transferability: local training versus cross-city transfer scenarios for UNetFormer, Model-D, and SantNet.

As shown in Table SV (see the Supplementary Material), when trained on the clean labels, SatLas achieved the highest OA (94.55%), and both foundation models outperformed SantNet in all per-class $F1$-scores, confirming their strong adaptability to downstream tasks. However, once label noise was injected, their performance deteriorated significantly, with OA dropping by over 10%. GFM experienced the largest decline, with OA decreasing by 16.82%, while Satlas also suffered a substantial drop. In contrast, SantNet incurred a much smaller accuracy loss under the same noisy dataset due to the explicit noise-handling mechanism, demonstrating superior robustness to mislabeled training data.

F. Model Complexity

To evaluate the computational efficiency and complexity of SantNet, we compared it with three comparative methods: UNetFormer, MANet, and HRNet. The results are summarized in Table SVI. SantNet contains fewer parameters than UNetFormer and MANet while having a slightly higher parameter count than HRNet, which also serves as SantNet's backbone.

In terms of training computational complexity, MANet is the most efficient model, showing the lowest computation and time cost. By contrast, SantNet requires approximately three times the computation and training time of HRNet. A detailed breakdown reveals that the additional overhead mainly arises from NC and semantic augmentation. The former requires frequent retrieval of previous prediction records, intensifying I/O operations, whereas the latter relies on extensive sampling to construct contrastive pairs, thereby increasing computational load.

Despite the higher training expense, the time-consuming modules in SantNet can be detached during inference, greatly reducing the computational cost.

G. EULCC Mapping

To assess the effectiveness of the proposed method in practical applications, we conducted fine-scale land cover mapping for urban areas in CONUS. Fig. 11 exemplifies the EULCC results of several cities, which delineate the heterogeneous distribution of both man-made features and natural features in detail under complex urban contexts.

To validate the accuracy of EULCC-CONUS, we randomly sampled 250 points in each state of CONUS and conducted visual interpretations by interpreters with domain expertise. In total, we constructed a nationwide pixel-level validation set containing 12243 samples across the CONUS for accuracy assessment. Using this validation set, we measured the confusion matrix of the proposed product, as shown in Table IV. Overall, EULCC-CONUS achieved an OA of 88.77%. From the perspective of producer's accuracy, *Water* class performed the best at 95.60% and then followed by *Tree*, *Road*, *Building*, *Low vegetation*, and *Parking*. *Others* and *Agricultural* achieved the lowest accuracies. Specifically, the spectrum of agricultural land is similar to that of grass, shrub, or bare land, leading to about one-third of *Agricultural* validation samples being wrongly classified as *Low vegetation*. Due to the complexity and vagueness of *Others* category, the pixels in this class are confused with many categories, including impervious land cover (i.e., *Road*, *Building*, and *Parking*) and natural land cover (i.e., *Low vegetation* and *Water*). In terms of user's accuracy, all classes achieved accuracies of over 80%, with

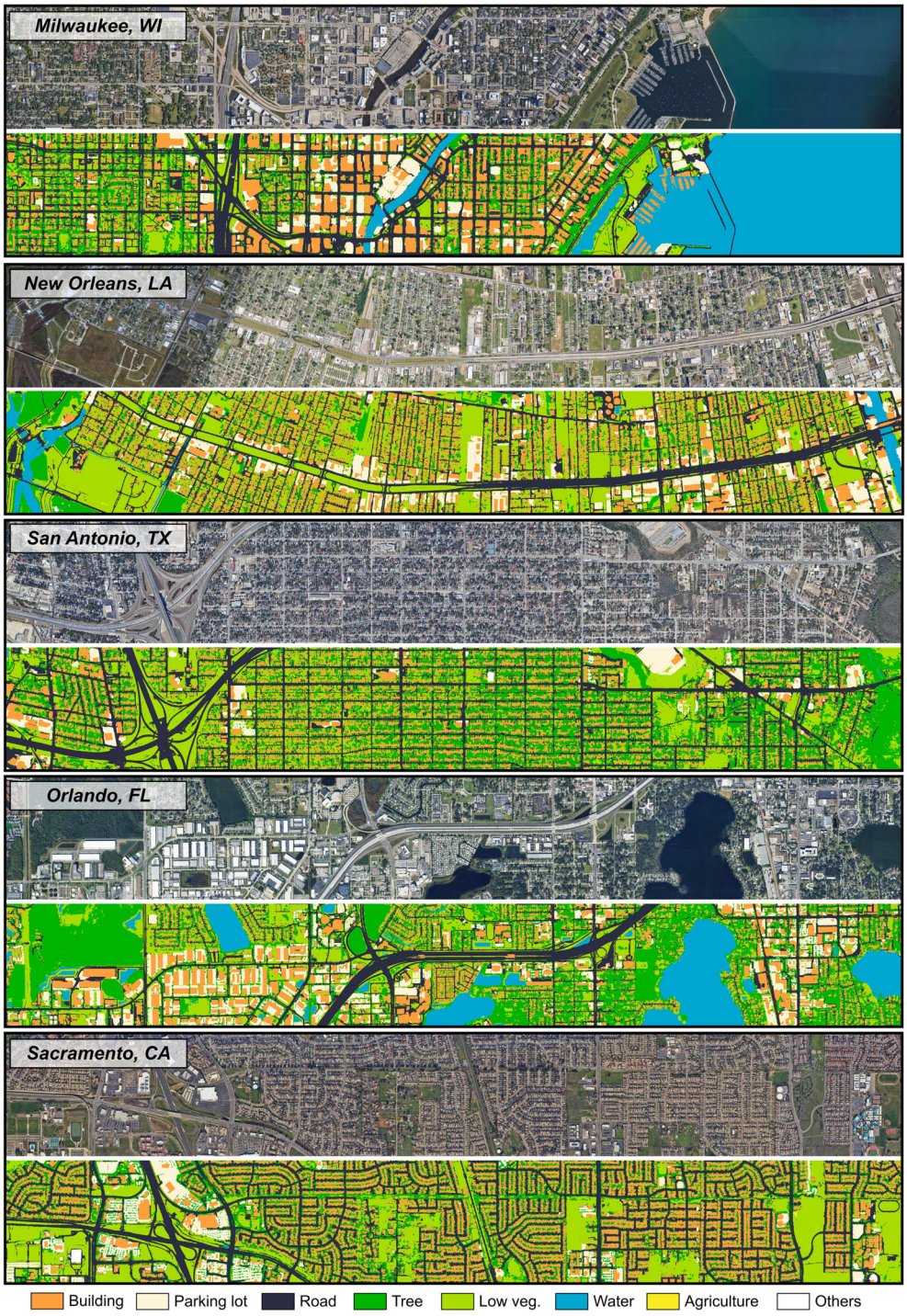


Fig. 11. Visualization of land cover maps for five sample cities from the generated EULCC product.

Others class having the lowest accuracy of 80.85%. The spatial distribution of state-level OA is illustrated in Fig. 12. Most states achieve OA above 80%. The dataset performed worse in the Midwestern states compared to other states, with New Mexico, Nevada, Wyoming, and Nebraska having relatively low OA. Among them, New Mexico has the lowest accuracy with the OA of 64%.

In addition to the accuracy assessment, we visually compared EULCC with existing regional or global land cover products within urban areas to demonstrate the characteristics of the generated product. Fig. 13 compares three regional high-resolution land cover products: UrbanWatch (2014–2017), EarthDefine (2018–2020), and Chesapeake Bay Watershed (CBW, 2017/2018) land cover datasets. Given that these regional products have varying coverage areas, two cities were selected from each product for a detailed comparison. For UrbanWatch, which has been used as labels in model training, the comparison clearly demonstrates that our products

TABLE IV

CONFUSION MATRIX FOR THE EULCC PRODUCT USING THE VALIDATION SAMPLE SETS COLLECTED ACROSS CONUS. NOTE THAT PA REPRESENTS THE PRODUCER'S ACCURACY AND UA REPRESENTS THE USER'S ACCURACY

Reference	*Building*	*Road*	*Parking*	*Tree*	*Low veg.*	*Agriculture*	*Water*	*Others*	Total	PA (%)
Building	1098	76	17	23	45	4	5	1	1269	86.52
Road	21	1537	8	14	49	1	6	0	1636	93.95
Parking	35	82	465	6	23	0	3	3	617	75.36
Tree	7	10	0	3803	152	0	11	0	3983	95.48
Low veg.	10	136	21	398	3437	16	22	5	4045	84.97
Agriculture	0	8	0	3	109	186	2	0	308	60.39
Water	1	3	1	3	6	0	304	0	318	95.60
Others	6	10	5	0	7	0	1	38	67	56.72
Total	1178	1862	517	4250	3828	207	354	47	12243	
UA (%)	93.21	82.55	89.94	89.48	89.79	89.86	85.88	80.85		
OA (%)	88.77									

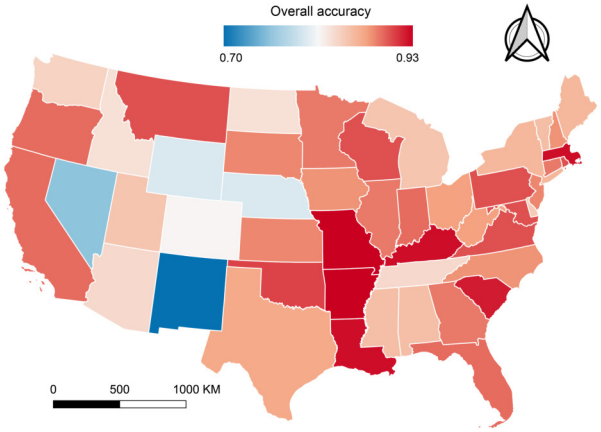


Fig. 12. State-level geographical distribution of the product's OA.

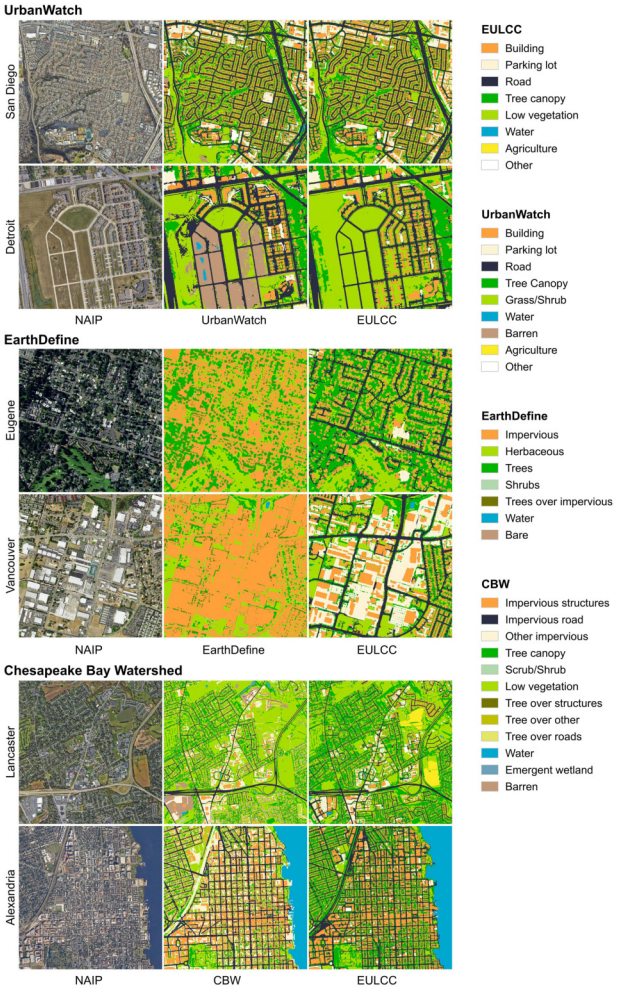


Fig. 13. Comparisons between EULCC and three high-resolution land cover products, including UrbanWatch, EarthDefine, and CBW land cover maps.

did not simply replicate the incorrect distributions present in UrbanWatch, most of which arise from land cover changes. SantNet corrected irregular features and detected changes in land objects due to temporal differences, such as the grove in the lower left corner of the San Diego scene as well as the newly constructed buildings in the center of the Detroit scene.

The land cover distributions in EarthDefine and EULCC are highly consistent, especially impervious surfaces and vegetation. The advantage of EULCC over EarthDefine is that more fine-grained impervious categories were differentiated, providing more detailed delineation of built environments. The CBW dataset, generated semi-automatically from aerial images, light detection and ranging (LiDAR) data, and Geographic Information System (GIS) layers, offers a high level of accuracy and confidence. In the Lancaster scene, the EULCC accurately updated the distribution of new impervious surfaces and classified agricultural land, a category not included in the classification system of CBW. In Alexandria, EULCC detected more buildings but fewer other impervious surfaces than CBW. Discrepancies exist in classifying the same land objects between the two datasets. For example, railways are classified as other impervious surfaces in the CBW, but as roads in EULCC. A similar situation occurs with informal roads and sidewalks.

We further compared EULCC in Orlando with four coarse-resolution land cover datasets: 10-m ESA Worldcover, 10-m

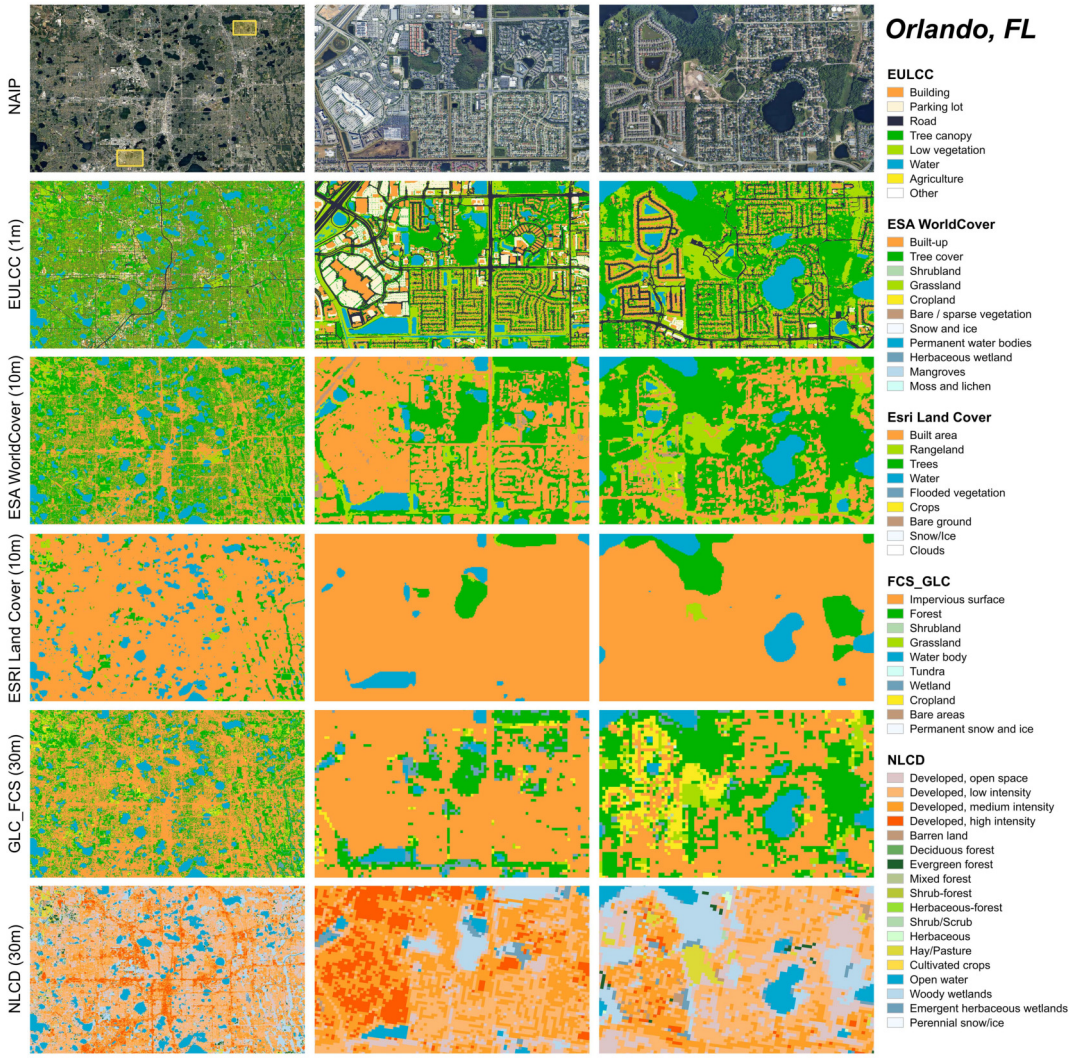


Fig. 14. Comparisons between EULCC and four medium-resolution land cover products, including ESA WorldCover, ESRI Land Cover, GLC_FCS, and NLCD.

ESRI Land Cover, 30-m GLC_FCS, and 30-m NLCD. The comparative results are illustrated in Fig. 14, exhibiting the superior performance of our product in capturing land objects of varying scales and preserving fine details. For large-scale land cover such as the widespread lakes in the example region, EULCC shows high consistency with other products while offering more precise shape and extent. The strength of our product also lies in the detailed delineation of urban features. Among the two 10-m resolution datasets, the Worldcover dataset can capture the basic distribution of impervious surfaces within cities to some extent, while the ESRI dataset seriously overestimates the built-up areas, overlooking urban green spaces. Due to the limitation of coarser resolution, two 30-m products cannot reveal urban land cover patterns with the same level of details and granularity as EULCC or Worldcover, although they still perform better than the ESRI product. The NLCD dataset, with its detailed impervious categories representing different levels of development intensity, provides better accuracy and detail in describing urban landscapes compared to GLC_FCS.

V. Discussion

To mitigate the negative impact of label noise on model training, two common strategies are typically considered: 1) reducing the generation of label noise at the source and 2) adapting the model to noisy labels, enabling it to leverage potential information from corrupted datasets. The second strategy is the main focus of this study. To cope with the adverse effect of potential noise from public land cover products, we integrated the NHM into deep learning networks. Additionally, an enhanced backbone (HPB) and a noise-adapted contrastive learning module (SAM) were incorporated to improve the mapping performance.

Under the same simulation settings, the proposed model achieved superior performance, showing stronger robustness to noise compared to other benchmark models. Although the benchmark models achieved lower accuracies, as shown in

Table I, most did not collapse and still presented acceptable results even in harsh settings with over 40% of labels being incorrect, suggesting that deep neural networks inherently possess certain levels of anti-noise capability, maintaining reasonable performance amidst noisy labels. Nevertheless, the general model typically demonstrates robustness only during the early training stages. As training progresses, these models initially show an increase in accuracy but then begin to decline dramatically after reaching a certain epoch, similar to trends observed in our ablation studies. Therefore, optimal parameters for these comparative models are often obtained in the initial training stages, making it crucial to have an accurate, clean validation set; otherwise, reliability may be compromised. In practical mapping applications, it is therefore indispensable to make additional efforts to collect precise validation samples, especially when models are trained for extensive study areas. In contrast, our model not only demonstrates better performance in evaluation metrics but also greater stability during later training stages. Its performance consistently improved throughout the training period and eventually reached a higher plateau. Even if the accuracy no longer increases, it does not decline significantly like other comparative models, ensuring reliable performance. This implies that even without a separate validation set, our model guarantees that the performance in the later stages surpasses earlier stages, guaranteeing the significance of learning efforts and associated satisfactory accuracy of the final well-trained model.

Previous studies have examined the robustness of mapping methods to varying levels of noisy labels. Gong et al. [76] found that the random forest classifier maintains stable accuracy—defined as within 1% variation—under the circumstance of up to 20% pixel-level noise. Similarly, Zhang et al. [36] also maintained stability with no more than 16% label noise. For pixel-level noise, deep learning methods, such as Dong et al. [45] and our study, show that accuracy decline remains within 1% across a broader range of label noise, indicating stronger resilience of deep learning to pixel-level noise. The studies by Zhang et al. [36] and Gong et al. [76] primarily focused only on pixel-level label noise, as their methods are pixel-based and minimally consider the neighboring pixel information. For deep learning-based semantic segmentation tasks, it is also important to test the robustness against object-level noise, as explored in [45] and our study. In these methods, the classification of a single pixel is influenced not only by its own features but also by the surrounding contextual information. Pixel-level noise can be smoothed and mitigated through these contextual relationships, whereas object-level noise remains more challenging and leads to serious performance degradation. In high-resolution land cover mapping tasks, object-level noise represents more general and potential label errors, such as misclassification of object categories during annotation or dynamic changes in land cover objects, while isolated pixel-level noise, such as salt-and-pepper noise in label, occurs less frequently. Therefore, model compatibility to handle object-level noise is imperative. SantNet's exceptional robustness is primarily attributed to the NHM module, allowing low coupling and high flexibility. This module can be integrated into existing architectures without modifying the core segmentation model, requiring only additional processing operations at the discrimination layer. This plug-and-play nature could allow the NHM to be easily implemented in other models, enhancing their resilience to label noise.

Using SantNet and a public land cover dataset, we generated a high-accuracy, high-resolution urban land cover product of the CONUS, accurately detailing the spatial patterns of essential urban features, such as buildings, roads, parking lots, vegetation, and water bodies. This country-scale dataset is expected to support a broad range of urban applications. For example, the accurate delineation of different impervious surface types contributes to building the foundational urban geographic information system and driving the advancement of digital twin cities. On the other hand, the precise positioning of natural components within urban areas, which are often underestimated in most large-scale land cover products, could benefit a variety of urban environmental assessment, such as green space supply-demand relationship [77], [78], urban heat island mitigation [79], [80], urban carbon accounting [81], ecosystem service evaluation [82], and public health and well-being promotion [83], [84]. Beyond its utility for diverse urban applications, this highly accurate—though not flawless—land cover product can serve as a valuable benchmark for further land cover mapping studies designed to enhance stability and noise resilience. We plan to continuously update the EULCC dataset as new NAIP imagery and improve modeling techniques become available. This will allow the construction of consistent and temporally extended land cover maps, enabling more accurate characterization of urban land cover dynamics. Meanwhile, we also encourage users to conduct local-scale accuracy assessments by collecting localized validation samples in regions of interest prior to application. This can be done by comparing it with authoritative datasets, field survey points, or through visual inspection. Such community-driven validation can enhance the product's confidence and motivate further improvements to our mapping framework.

Although SantNet exhibits superior mapping performance and robustness to noisy labels, several limitations remain. First, integrating three improvement modules increases computational burden, especially during training. However, except for the high-resolution preserving branch, the NHM and SAM modules operate only during training and can be totally removed during inference, reducing computational costs. Future work can consider incorporating lightweight structures into the high-resolution backbone to improve the model efficiency [85], [86].

Second, the hyperparameter settings influence classification performance, limiting framework adaptability and scalability. Future work could explore more automated and adaptive strategies to enhance the flexibility of our methods. For example, these hyperparameters could be formulated as learnable parameters and jointly optimized with network weights in a data-driven manner.

Third, regarding the generated EULCC product, the *Barren* class was merged with grassland due to the low accuracy and high confusion with grassland. Besides, *Agricultural* and *Other* classes also have less accurate classification. To improve this, thematic mapping for each class can be specifically

carried out [87]. For *Agricultural* land, time series imagery may help improve the classification of these land cover types due to the significant characteristics of dynamic change over time [88], [89]. The *Other* class defined by Zhang et al. [54] actually involves a broad range of land cover types such as artificial turf, acrylic surface, transmission towers, and construction sites. Classifying these land cover types into one class is challenging due to their distinct characteristics. Therefore, more detailed classes can be separated from the *Other* class to construct a more complete and fine-grained urban land cover classification system in the future. To accurately distinguish these fine-grained classes, multisource geospatial data such as various kinds of social sensing data and other remote sensing data can be integrated for the improvement of classification performance [90].

VI. Conclusion

This study developed a novel deep learning-based framework for urban land cover mapping with noisy labels, named SantNet, which incorporates three unique features: HPB, NHM, and SAM. The HPB effectively maintains the high-resolution representation throughout the network, enhancing the spatial details of land cover objects. The NHM ensures robustness against erroneous labels, while the noise-adapted SAM significantly improves feature representations by leveraging contrastive learning under noisy labels. Extensive experiments verified the superior performance of SantNet in high-resolution land cover classification compared to other benchmark methods, highlighting its potential for large-scale mapping practices using publicly available but imperfect land cover products as references. As a practical application, we employed the SantNet to generate the EULCCs, 1-m resolution urban land cover maps covering the urban areas of CONUS. The EULCC achieved satisfactory OA, validated by over 20 000 samples. Compared to existing land cover products, EULCC provides the latest, higher resolution, finer granularity, and more detailed information on urban land cover patterns, serving as important data support for a wide range of urban applications. Looking ahead, developing a lightweight architecture and an efficient noise-handling design is expected to further improve the computational efficiency, while integrating multisource data and thematic mapping could potentially improve the performance of classes with relatively low accuracy.

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